

Fatigue is associated with stimulus-related theta and alpha power and distinguishes subjective from objective cognition in a post-viral cohort: A cross-sectional study

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Abstract

Background

Fatigue is the main driver of impaired quality of life in post COVID-19. Yet, fatigue overlaps with depression, anxiety, poor sleep, and cognitive complaints. Task-related neurophysiological correlates have been largely unexplored and subjective symptoms often diverge from objective cognitive performance. We investigated whether fatigue (i) is independently associated with task-related cognitive performance, (ii) has identifiable EEG correlates, and (iii) mediates the relationship between subjective cognitive symptoms and objective performance.

Methods

In 98 individuals with prior PCR-confirmed SARS-CoV-2 infection from the COVIDOM/NAPKON study in Kiel, we recorded 128-channel EEG during a psychomotor vigilance task. Fatigue (FACIT-Fatigue subscale), depression and anxiety (HADS), and sleep quality (PSQI) were assessed. Partial correlations, group comparisons of pre- versus post-stimulus theta and alpha peak power, and a bootstrap mediation analysis were performed. Key correlations were replicated in the population-based COVIDOM sample from Kiel ($n = 2,549$). We investigated the oscillatory activity by removing aperiodic components from the total signal.

Results

Stronger fatigue was independently associated with slower reaction times, whereas depression, anxiety, and sleep were not. This finding was replicated in the larger COVIDOM sample from Kiel. The increase in theta and alpha peak power from the pre- to the post-stimulus period was significantly smaller in participants with clinically significant fatigue. Descriptively, these participants showed slightly higher pre-stimulus theta and alpha power and lower post-stimulus peak power than participants without clinically significant fatigue. The presence of subjective cognitive symptoms (disorientation, slowed thinking, difficulties concentrating, or forgetfulness) was associated with slower reaction times. This association was substantially mediated by fatigue (39% of the effect), acknowledging potential measure overlap.

Conclusions

Fatigue is a construct with distinct behavioral and electrophysiological signatures and a central link between subjective cognitive complaints and objective deficits, suggesting fatigue as an essential target for clinical assessment and intervention.

Introduction

Fatigue

Fatigue is defined as the sensation of depleted energy reserves accompanied by an increased need for rest that is disproportionate to the preceding exertion [1]. It is a major symptom of various conditions, including post COVID-19 condition (PCC) [2], chronic fatigue syndrome [3], multiple sclerosis [4], and cancer [5]. Although healthy individuals can also experience fatigue, 'clinically significant fatigue' (CSF) is characterized by high severity, significant impairment of daily activities, and persistence over time (often without improvement through rest), features that distinguish CSF from other forms of everyday fatigue [3, 6, 7].

Given the consensus that fatigue is a subjective sensation, it has consequently been measured to date only by questionnaires or clinical interviews [8, 9]. In contrast, the objectively measurable, temporary decline in motor or cognitive performance induced by activity during a task is referred to as 'fatigability' [8, 10]. Fatigue and fatigability are considered distinct constructs, as the correlations between them are generally quite weak [8, 11]. Nevertheless, individuals who report higher levels of fatigue also tend to exhibit greater fatigability [12].

Fatigue has been identified as the key factor impairing quality of life after infection with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) [13]. It was thus found to occur significantly more frequently in individuals following an infection than in a randomized, Germany-wide control cohort assessed prior to the pandemic [14]. Structural alterations in subcortical brain regions have been proposed as a cause of the severity of PCC-associated fatigue [15]. It is noteworthy that fatigue frequently co-occurs with depressive symptoms, poor sleep quality, cognitive symptoms, and anxiety [16, 17]. However, since these conditions are strongly correlated [18, 19], it often remains unclear whether fatigue is a primary symptom of a specific health issue or merely a consequence of other symptoms. Therefore, the assessment of fatigue always requires careful consideration of, and differentiation from, potentially overlapping affective and cognitive symptoms. A simple reaction-time task that is widely used in research on sleep and fatigue is the PVT [20–22].

Electrophysiological correlates

Electroencephalography (EEG) signals reflect the summed postsynaptic activity of (sub)cortical neurons, comprising both periodic (oscillatory) and aperiodic activity [23]. Thus, the power spectral density not only encompasses periodic oscillations such as in the alpha and theta frequency band, but also has a distinct aperiodic component characterized by a 1/f-like distribution, reflecting background neuronal activity [24, 25]. Since distinct physiological correlates of aperiodic activity have been identified, periodic and aperiodic components should clearly be parameterized and analyzed separately [26]. This separation is important in practice because changes in the aperiodic component can influence the measurement of oscillatory activity and can thus lead to misinterpretation of the signal [24, 27].

Previous research has aimed to identify electrophysiological signatures of fatigue mainly in resting-state EEG data, with increased theta power and decreased alpha power identified as the most promising candidates [28, 29]. In addition to resting-state EEG, task-related EEG data show that both frontal/frontomedial theta power and parietal alpha power increase over the course of a task, accompanied by increasing error rates and prolonged reaction times [30–34]. These changes have been interpreted as a compensatory effort in response to growing mental fatigue and declining attentional performance [30, 33, 34]. An increase in task-related theta power is associated with early preparatory processes involved in attention and memory encoding, and can therefore occur during or prior to stimulus presentation [35]. Similarly, higher pre-stimulus alpha power has been found to be associated with reduced sustained attention and, hence, to predict subsequent errors [36]. However, although they claimed to investigate mental fatigue, task-related EEG studies have merely examined fatigability, while the task-related neurophysiological correlates of fatigue have remained largely unexplored. An especially understudied area is the relationship between fatigue and brain activity after the presentation of a task-relevant stimulus ('post-stimulus').

Other studies investigating cognitive function independently of fatigue found that higher task-related theta power during cognitive processing and lower alpha power during task performance are both associated with better cognitive performance and less cognitive impairment [37, 38]. Furthermore, activity in the theta frequency band is a key component of the P3 wave [39], the amplitude of which is consistently reduced in individuals with lower cognitive functioning and greater fatigue [40–46]. Likewise, post-stimulus alpha power has been shown to be associated with cognitive performance and P3 amplitude [47].

Brain fog and cognitive symptoms

Patients with various medical conditions frequently self-report a sensation known as 'brain fog', characterized by deficits in cognitive performance, fatigue, and affective symptoms [48]. However, subjective cognitive symptoms and objective cognitive performance are not strongly correlated with each other and are therefore typically regarded as distinct concepts [48–50]. For instance, most individuals with persistent symptoms following SARS-CoV-2 infection report cognitive symptoms and fatigue [51, 52], but although their performance on neuropsychological test is poorer on average, their subjectively perceived cognitive symptoms often do not align their actual cognitive performance [49, 53, 54]. In fact, identifying the reasons for subjective cognitive symptoms associated with brain fog is challenging because these symptoms can stem from a variety of causes, including fatigue, depression, sleep disturbances, or structural brain changes [15, 55–57]. Anyhow, there are good reasons to assume that subjective cognitive symptoms are primarily linked to fatigue [49], as the latter appears to be the key mediator between subjective cognitive symptoms and cognitive performance [58].

Research question

The aim of the present study was to investigate in detail the interrelationships between fatigue and task-related cognitive performance as well as their underlying neurophysiological correlates. We

hypothesized that fatigue, but not depressive symptoms, anxiety or sleep quality, was independently associated with task-related cognitive performance. To examine the role of neurophysiological activity in fatigue, we compared the task-related EEG profiles of individuals with and without CSF, defined by a validated questionnaire cutoff. In doing so, we focused upon theta- and alpha power before and after stimulus onset, as changes in both frequency bands have consistently been linked to fatigue before. Finally, we hypothesized that individuals who reported cognitive symptoms perform worse on a psychomotor vigilance task than those without such symptoms, and that this difference could be explained by fatigue. The study was conducted on a cohort of individuals with a prior SARS-CoV-2 infection and, consequently, a high prevalence of persistent fatigue.

Methods

Study cohort

The participants of the present study (N = 98) were randomly recruited as a sub-cohort of COVIDOM in Kiel (also referred to as 'NAPKON-POP'). COVIDOM is a large, longitudinal multicenter study embedded within the National Pandemic Cohort Network (NAPKON), with study sites in Berlin, Kiel, and Würzburg [59]. Every COVIDOM participant who took part in the follow-up examination at the Kiel study site between June 2023 and July 2025 was eligible for the present study. These individuals were ≥ 18 years old, resided in Schleswig-Holstein and, like all other COVIDOM participants, had a prior SARS-CoV-2 infection diagnosed by Polymerase Chain Reaction (PCR) test [59]. Both in COVIDOM and in the present study, written informed consent was obtained from all participants prior to their participation, in accordance with the Declaration of Helsinki. The present study was approved by the ethics committee of Kiel University (reference number: Kiel D446/23) and registered in the German Clinical Studies Registry (ID: DRKS00032132; Registration Date: 26 June 2023).

Study procedures

The COVIDOM anamnesis interview, among other things, also asked for neurological symptoms that had developed either during the acute phase of COVID-19 or afterwards. One response option addressed the presence of cognitive symptoms, including disorientation, slowed thinking, difficulties concentrating, or forgetfulness, which hereafter will be referred to as 'subjective cognitive symptoms'. In addition, COVIDOM participants underwent cognitive screening using the Montreal Cognitive Assessment (MoCA), where total scores < 26 are considered indicative of cognitive impairment [60].

All participants of the present study paid a one-time study visit to the Department of Neurology, University Hospital Schleswig-Holstein, Campus Kiel, where they first performed the paper-and-pencil Trail Making Test [61]. Subsequently, an EEG was performed comprising recordings during four computerized cognitive tasks as well as a resting-state measurement. Finally, participants were asked to complete three questionnaires to assess fatigue, depressive symptoms, anxiety symptoms, and sleep quality.

Cognitive performance and fatigue assessment

All four computerized cognitive tasks performed during EEG recording had been programmed in PsychoPy (version 2022.2.4) and presented to the participants on a 27-inch computer screen placed at a distance of 1 meter. For the purpose of the present study, we confined the analysis to the Psychomotor Vigilance Task (PVT) [62] because it efficiently captures the cognitive domains of sustained attention and alertness in the reaction time.

Each trial started with the presentation of a white fixation cross for 2–10 seconds, followed by the display of a red number, counting the milliseconds elapsed since its onset (Fig. 1). Participants were instructed to press the spacebar as quickly as possible as soon as the red number appeared. Following this response, the counter stopped and displayed the individual reaction time for 500 ms. Then, the fixation cross reappeared to initiate the next trial.

Fatigue was assessed using the Functional Assessment of Chronic Illness Therapy – Fatigue Subscale (FACIT-Fatigue subscale) questionnaire [63], which was generally employed throughout the COVIDOM study [16]. The questionnaire addresses the presence of self-reported fatigue, its severity, and its impact on daily activities and functioning over the preceding 7 days [64]. A resulting score ≤ 30 is usually considered indicative of CSF [65]. Anxiety and depression were assessed with the Hospital Anxiety and Depression Scale (HADS) [66], a self-report instrument developed for use in somatic settings that deliberately excluded symptoms that also occur in physical disorders (e.g., fatigue, headaches). The German translation of the HADS was used in the present study [67]. Notably, the HADS score can be split into an anxiety sub-score (HADS-A) and a depression subscore (HADS-D). Sleep quality was assessed with the Pittsburgh Sleep Quality Index (PSQI) [68]. The PSQI refers to the sleep behavior of the last four weeks.

EEG recording

EEG was recorded using a 128-channel cap with equidistant electrode layout (128Ch Standard Brain Cap for actiCHamp Plus, Easycap GmbH, Wörthsee, Germany) and an actiCHamp Plus Amplifier (Brain Products GmbH, Gilching, Germany). The sampling rate was 1000 Hz with an amplitude resolution of 0.1 μ V. Electrode impedances were kept below 20 k Ω . The reference electrode was placed on the nose, and two electrooculogram electrodes were positioned diagonally beneath each eye.

EEG preprocessing

EEG data were analyzed using Matlab R2024b (version: 24.2.0.2806996) [69] with custom-written scripts based upon the Fieldtrip (version: 20250106) [70] and EEGLAB (version: 2024.2) [71] toolboxes.

Preprocessing was carried out in two streams:

1. For artifact identification, the continuous data underwent DC correction and band-pass filtering (1 to 40 Hz; onepass-zero phase, Hamming-windowed sinc Finite Impulse Response), followed by the removal of noisy channels based upon variance criteria (sample median: 39 removed channels,

- range: 0–48), and of 1-second segments with $|z| > 2$ (sample median: 23 removed segments, range: 0–36). Subsequently, independent component analysis (*runica*, EEGLAB) was performed, with components automatically classified as muscle, eye, heart, line noise, channel noise, or other artifact using *ICLabel* [72] and *pop_icflag* (sample median: 58 removed components, range: 0–86).
2. For spectral power analysis, the raw data were band-pass filtered (1 to 90 Hz) and a 50 Hz band-stop was applied. After noisy channels and artifactual components identified in stream 1 had been removed, the data were segmented into 2.4-second epochs spanning ± 1.2 seconds around each PVT stimulus onset. Trials with extreme joint probability were excluded (sample median: 4 removed trials, range: 1–11), removed noisy channels were interpolated using spherical splines, and the signal was re-referenced to common average. For complete filter specifications and rejection criteria, see Supplementary Methods.

EEG spectral power analysis

EEG spectral power was computed separately for the 1-second pre- and post-stimulus intervals in order to distinguish event-related and non-event-related activity [73]. Trial-averaged power spectra (1 to 40 Hz) were computed by way of fast Fourier transformation using the *mtmfft* method in Fieldtrip with 4 seconds padding and a *dpss* taper providing 4 Hz smoothing. The aperiodic component was computed using the *FOOOF* algorithm in *knee* mode [74] and was then subtracted from the total spectrum to obtain the oscillatory component, averaged over channels. Alpha peak power was defined as the maximum power within 8 to 12 Hz whereas theta peak power was defined as the maximum power within 4 to 7 Hz.

Statistical analyses

All data preparation, visualization, and statistical analyses were performed with R (version: 4.4.2) [75] and JASP (version: 0.19.3.0) [76]. EEG-specific data processing and visualizations (topographical plots) were done with Matlab R2024b [69]. All scripts and data used in the present study are publicly available at github.com/christianneumannCAU/EPOC-Kiel and openneuro.org/datasets/ds007605, respectively.

The study cohort was divided into two sub-groups according to the CSF cutoff of the FACIT-Fatigue subscale, and demographic data were summarized separately for each group. Differences in age, years of education, and MoCA total score were assessed for statistical significance using a two-tailed Mann-Whitney U test for independent samples. Differences in the distribution of gender and pre-diagnosed neurological disorders between groups were analyzed with a χ^2 tests of independence. *P* values < 0.05 were considered statistically significant throughout.

Association between fatigue and task-related cognitive performance

To assess the level of association between fatigue and task-related cognitive performance after adjustment for depression, anxiety, and sleep quality, a partial Spearman rank correlation analysis was carried out between FACIT-Fatigue subscale score and PVT reaction time, controlling for the following covariates: HADS-D total score, HADS-A total score, PSQI total score, age and years of education. To

examine whether depression, anxiety, or sleep quality are independently associated with cognitive performance, a partial correlation analysis was also performed between PVT reaction time and HADS-D total score, HADS-A total score, and PSQI total score, respectively, controlling for the same psychological and demographic covariates. *P* values were Holm-corrected to account for the multiplicity of these four primary correlation analyses.

The partial correlation analyses were then repeated for the complete COVIDOM cohort from Kiel ($n = 2,549$). Here, PSQI and years of education were not included as control variables because the data were not available. Moreover, depressive symptoms and anxiety were measured in COVIDOM by the *Patient Health Questionnaire – 8* (PHQ-8) [77] and the *Generalized Anxiety Disorder 7-item scale* (GAD-7) [78].

Neurophysiological correlates of fatigue

Pair-wise differences between the pre- and post-stimulus measurements of theta (4 to 7 Hz) and alpha (8 to 12 Hz) peak power were tested for statistical significance using a two-tailed Wilcoxon signed-rank test. Whether these differences were significantly smaller in individuals with CSF than without CSF was assessed using an one-tailed Mann-Whitney U test. *P* values were Holm-corrected for the two primary tests.

Role of subjective cognitive symptoms

Participants in the present study were also stratified according to their answer (“yes” or “no”) to the COVIDOM neurological anamnesis question about symptoms developed in connection with COVID-19 (see above). Resulting group differences in PVT reaction time were assessed for statistical significance using a one-tailed Mann-Whitney U test, and the effect size was quantified by the corresponding rank-biserial correlation. In addition, the effect of fatigue on the association between subjective cognitive symptoms and PVT reaction time, was assessed by regression-based mediation analysis, using the *mediation package* [79] of R. Indirect and direct effects were estimated by nonparametric bootstrap resampling (10,000 simulations), controlling for HADS-D total score, HADS-A total score, and PSQI total score, age and years of education. Holm correction was applied to *p* values of the two primary mediation tests.

Results

Study characteristics

Instances of missing data were noted, with some overlap, for PVT ($n = 1$), MoCA ($n = 1$), FACIT-Fatigue subscale ($n = 2$), HADS-D total score ($n = 3$), HADS-A total score ($n = 1$), PSQI total score ($n = 6$) and EEG recordings ($n = 3$). Reasons included time and health-related constraints, failure to record button responses, missing key questionnaire items, and corruption of data files. All behavioral data fell within expected ranges, and no outliers were removed. Upon visual inspection, EEG peak power values

exceeding the mean by at least 5 *SD* were classified as outliers, resulting in the exclusion of one participant from subsequent EEG analyses.

No statistically significant difference was observed between study participants with ($n = 37$) and without CSF ($n = 59$) in terms of age ($U = 1210.5, p = 0.37$), years of education ($U = 979.5, p = 0.40$), gender ($\chi^2(1, 96) = 0.95, p = 0.33$), pre-diagnosed neurological disorders ($\chi^2(1, 96) = 1.98, p = 0.16$), or MoCA total score ($U = 947.5, p = 0.33$) (Table 1).

Table 1
Sociodemographic and clinical characteristics of study participants.

Clinically significant fatigue (CSF)	N	Age in years (Median, IQR)	Female (%)	Years of Education (Median, IQR)	Pre-Diagnosed Neurological Disorders (%)	MoCA total score (Median, IQR)
Yes	37	55 (45–57)	25 (68)	15 (13–16)	13 (35)	27 (26–28)
No	59	52 (35–59)	34 (58)	15 (13–18)	13 (22)	28 (26–29)

Association between fatigue and task-related cognitive performance

Stronger fatigue (i.e., a lower FACIT-Fatigue subscale score) was significantly associated with slower reaction times in the PVT ($\rho = -0.29, p = 0.03$) after adjustment for HADS-D total score, HADS-A total score, PSQI total score, age and years of education. No significant correlation with PVT reaction time was observed for HADS-D total score ($\rho = -0.13, p = 0.77$), HADS-A total score ($\rho = 0.09, p = 0.88$), or PSQI total score ($\rho = -0.07, p = 0.88$).

Similar results were obtained in the complete COVIDOM cohort from Kiel. Slower PVT reaction times were associated with stronger fatigue ($\rho = -0.11, p < 0.001$) after adjustment for PHQ-8, GAD-7 and age, but not with PHQ-8 ($\rho = 0.03, p = 0.18$) or GAD-7 ($\rho = -0.03, p = 0.24$) itself.

Neurophysiological correlates of fatigue

Both alpha ($W = 120, p < 0.001$) and theta ($W = 39, p < 0.001$) peak power increased significantly from pre-stimulus to post-stimulus during the PVT (Fig. 2). The increase was significantly smaller in participants with CSF than in those without CSF, both for theta ($U = 647, p = 0.002$) and alpha ($U = 703, p = 0.004$) peak power (Fig. 2). Moreover, participants with CSF exhibited slightly higher pre-stimulus theta and alpha power, on average, but lower post-stimulus peak power than participants without CSF. Topographical plots support these findings (Fig. 3–4): Across all participants, theta power predominantly increased over frontal albeit widely distributed regions while alpha power primarily increased over parietal regions. In addition, participants with CSF showed slightly higher pre-stimulus theta and alpha power in some regions. Consequently, the stimulus-related increase in power was more pronounced in participants without CSF than in those with CSF (Fig. 2–4).

Task-related cognitive performance and subjective cognitive symptoms

Participants reporting subjective cognitive symptoms had significantly slower PVT reaction times than those without such symptoms ($U = 1601, p < 0.001$), with a rank-biserial correlation of 0.44. Mediation analysis revealed a significant indirect effect ($p = 0.04$), but no significant direct effect ($p = 0.21$). The proportion mediated was 0.39, indicating that fatigue accounted for 39% of the association between subjective cognitive symptoms and PVT reaction time. After adjusting for fatigue, the direct association between subjective cognitive symptoms and PVT reaction time was no longer statistically significant. Overall, these findings were consistent with a substantial mediation of PVT reaction time by fatigue.

Discussion

In an observational cross-sectional study, we investigated the relationship between fatigue and task-related cognitive performance, as well as their electrophysiological correlates, among individuals with prior SARS-CoV-2 infection. We made three key findings:

First, stronger fatigue was associated with slower PVT-assessed reaction time, independently of potential mediators such as depression, anxiety, sleep quality, or age and education. This result was qualitatively replicated in the larger COVIDOM cohort from Kiel. Second, study participants with clinically significant fatigue (CSF) exhibited a smaller task-related increase in both alpha and theta peak power from pre-stimulus to post-stimulus period during the PVT than participants without CSF. This is also reflected descriptively by their higher average pre-stimulus and lower average post-stimulus theta and alpha peak power. Third, subjective cognitive symptoms, as defined by the COVIDOM anamnesis interview, were associated with slower PVT reaction times, and the association was substantially mediated by fatigue, considering possible content overlap of the self-report measures of fatigue and subjective cognitive symptoms.

Taken together, the above findings support fatigue as a distinct neuropsychological concept with measurable behavioral and electrophysiological qualities, and as a central common correlate of subjective cognitive symptoms and objective performance reduction after SARS-CoV-2 infection.

1. Fatigue is independently associated with task-related performance

Stronger fatigue was found to be correlated with slower PVT reaction times even after adjustment for depressive symptoms, anxiety, sleep quality, age and years of education. Notably, depressive symptoms, anxiety, and sleep quality themselves showed no significant association with PVT reaction time. Even though all of these symptoms are not easy to distinguish [17–19], our findings empirically confirm that fatigue, as a neuropsychological concept, is clearly distinct from overlapping affective and sleep-related symptoms. Our results thus suggest that fatigue has its own effect on sustained attention and alertness that cannot be reduced to comorbid affective disturbance. This is consistent with previous research showing that individuals with post-viral fatigue often experience cognitive impairments [55]. Replication of the observed relationship between fatigue and cognitive performance in a larger population-based sample confirms its robustness beyond ‘laboratory’ cohorts. This conclusion appears valid despite the use of different depression and anxiety assessment tools (PHQ-8, GAD-7) and a lack of PSQI data in the

larger sample. The smaller correlation coefficients noted in the latter were not unexpected, given the greater heterogeneity of COVIDOM. Importantly, however both the direction and the specificity of the effect were consistent.

The PVT is primarily used to study fatigue in the context of sleep deprivation and time-on-task effects, often measured by lapses [20, 22]. Our study takes this approach even further by demonstrating that the mean reaction time over a five-minute task is associated with fatigue in a way that is not sleep-related. Moreover, the association relates to subjective fatigue, i.e., a perception-level construct that is distinct from acute task-induced fatigability, consistently shown to be only weakly correlated with the former [8, 11, 12].

2. Neurophysiological correlates: reduced stimulus-related modulation

All participants showed an increase of both theta and alpha peak power from the pre- to the post-stimulus interval, with a frontal distribution observed for theta power and a parietal distribution for alpha power. This pattern is consistent with the established roles of the two neurophysiological characteristics in cognitive control and stimulus encoding (theta) [35, 80], and attention modulation (alpha) [81], respectively. Notably, participants with CSF exhibited a smaller pre- to post-stimulus increase in both bands due to a combination of higher pre-stimulus and lower post-stimulus peak power.

The observed association between CSF and higher pre-stimulus theta and alpha power is consistent with earlier reports of (i) an increased resting-state theta power in fatigued individuals [28, 29], (ii) increases in theta and alpha power over the course of a task accompanied by decreases in cognitive performance [30, 33, 34], and (iii) a link between higher pre-stimulus alpha power and reduced sustained attention [36]. The reported activities were observed in regions similar to those identified in the present study. These results suggest that individuals with fatigue are less well prepared for stimuli due to decreased attentional functioning. The descriptively higher average pre-stimulus theta and alpha power could also be interpreted as a compensatory mechanism to counteract increased effort.

While the reduced post-stimulus peak among study participants with CSF agrees with a previous observation of individuals with reduced cognitive function also exhibiting reduced task-related theta engagement [38], it contrasts with another report of higher alpha power during task performance in individuals with lower cognitive functioning [37]. However, both of these studies did not investigate the role of fatigue, rendering our study one of the first to reveal an attenuated increase in post-stimulus theta and alpha peak power in individuals with fatigue. The reduced theta and alpha peak increase in participants with CSF may reflect diminished stimulus processing due to fatigue, resulting in slower reactions.

Increased theta activity was observed between 200 and 400 ms post-stimulus (Supplementary Figure S1), a time window that corresponds to the N2 - P3 complex [73]. Frontocentral theta activity is considered a key component of P3 [39], the amplitude of which has been shown to decrease with prolonged reaction times and in individuals with post COVID-19 [40, 41, 44, 82].

In participants without CSF, theta activity appeared to be more widely distributed across the scalp and not confined to the frontal regions. This is consistent with the hypothesized role of theta oscillations as a communicator and coordinator between functionally relevant networks [80, 83, 84].

3. Subjective cognitive symptoms are explained by fatigue

Participants with self-reported cognitive symptoms showed slower PVT reaction times, with a moderate-to-large effect size (rank-biserial correlation = 0.44). However, mediation analysis indicated that this association was substantially mediated by fatigue. Indeed, 39% of the total effect was explicable by the influence of fatigue, and after adjusting for fatigue, the direct effect of subjective cognitive symptoms on reaction time was no longer statistically significant. Since depressive symptoms, anxiety, sleep quality, age, and years of education were held constant in the model, these factors could be excluded as possible alternative explanations.

Our result adds to a growing body of evidence that subjective cognitive complaints and formally measured cognition are only loosely linked [48–50, 53, 85], and that fatigue is a primary mediator between them [58]. Practically it suggests that, when a patient with prior SARS-CoV-2 infection reports “brain fog”, the most likely tractable clinical target and determinant of objective attentional performance is fatigue rather than cognitive symptomatology *per se*. Moreover, if fatigue is a primary mediator of self-reported cognitive symptoms, this would explain why such reports often do not align with task-based measures of cognitive performance in post COVID-19 [49], as impairments in sustained attention are expected to fluctuate over time [86].

The above distinction is crucial for clinical diagnostic and therapeutic decision-making in fatigue-related conditions such as post COVID-19. Cognitive training may initially appear to be a plausible intervention for patients reporting cognitive symptoms, as it directly targets cognitive function [87]. However, if fatigue rather than cognitive impairment *per se* is the main driver of these symptoms, cognitive training should be carefully reconsidered. An appropriate balance may be required in such cases to avoid overly demanding interventions that may exacerbate symptoms while offering limited benefit.

Limitations and future research

As the present study is cross-sectional in design, causal inferences about the direction of the relationship between fatigue and task-related cognitive performance cannot be drawn, although the result of the mediation analysis clearly suggests that fatigue is an upstream driver. Second, subjective cognitive symptoms were captured by a single interview item aggregating disorientation, slowed thinking, concentration difficulties, and forgetfulness. A more granular symptom inventory would allow finer-grained mediation modeling. In addition, some FACIT-Fatigue items describing the difficulty of starting or finishing usual activities may share experiential content with the aggregated cognitive symptom item, and the present study did not perform item-level discriminant validity checks. Third, all participants had a prior PCR-confirmed SARS-CoV-2 infection, and no non-infected control group was included. The results characterize the structure of fatigue and its correlates within this population but

cannot address whether the observed relationships are specific to post COVID-19 or can be generalized to fatigue in the context of other conditions. Fourth, the PVT captures sustained attention and alertness with high sensitivity but does not probe higher-order executive or memory processes. Whether the observed neural signature extends to other cognitive domains remains to be determined. Finally, the PVT was administered last in the study procedure. This could have exacerbated the association between fatigue and task-related cognitive performance due to higher fatigability.

In future research, alpha power could instead be analyzed using regions-of-interest approaches rather than whole-scalp averages, or be further decomposed to assess potential motor-related contributions. Given the strong association observed with fatigue over the preceding week, future studies should also consider longer temporal assessment windows.

Conclusions

In individuals with prior SARS-CoV-2 infection, fatigue is associated with slower reaction times during a psychomotor vigilance task independently of depressive symptoms, anxiety, and sleep quality. Clinically significant fatigue is accompanied by a reduced stimulus-related modulation of theta and alpha peak power, reflecting the combination of elevated pre-stimulus power and diminished post-stimulus peak. Subjective cognitive symptoms are associated with objective performance decrements, but this link is substantially mediated by fatigue and cannot be explained by the cognitive complaints themselves.

Taken together, our findings establish fatigue as a distinct construct with reproducible behavioral and electrophysiological correlates and as a key factor connecting subjective cognitive complaints to measurable attentional deficits after SARS-CoV-2 infection. Clinically, the results argue for assessing and prioritizing fatigue, rather than subjective cognitive symptoms alone, when patients report brain fog, and for weighing this distinction carefully when selecting interventions. Patients should be encouraged to consider how far fatigue drives their cognitive symptoms, and clinicians should take these complaints seriously.

Declarations

Ethics approval and consent to participate

In accordance with the Declaration of Helsinki, written informed consent was obtained from all COVIDOM participants and separately again for the present study. Both studies received ethical approval from the ethics committee (reference numbers: COVIDOM D537/20, and the present study D446/23). COVIDOM (DRKS00023742) and the present study (DRKS00032132) were registered in the German Registry of Clinical Studies, COVIDOM was also registered in ClinicalTrials.gov (NCT04679584).

Consent for publication

Not applicable.

Competing Interests

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Author Contribution

C.N., L.E., L.F., W.M., and J.W. contributed to the study conception and design. Material preparation was performed by C.N., L.F., S.P., L.R., T.B., W.L., S.S., M.K., J.H., and W.M. Data collection was performed by

C.N., L.E., L.F., L.S., M.B., J.M.H, K.H., K.F., S.P., J.G., A.R., L.S., S.R., J.J.V., and B.L. Analysis was performed by C.N., G.H., L.G., J.K. and J.W. The first draft of the manuscript was written by C.N., G.H., J.G., L.S., J.K., J.W. and W.M. All authors read and approved the final manuscript.

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Data Availability

The datasets generated and analysed during the current study are available in the openneuro repository, openneuro.org/datasets/ds007605. Scripts and information on how the data was processed are available at the github repository, github.com/christianneumannCAU/EPOC-Kiel. The COVIDOM datasets are available for collaborative research upon request via a transfer point of NAPKON (more information is available on the NAPKON website: <https://proskive.napkon.de/>).

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Figures

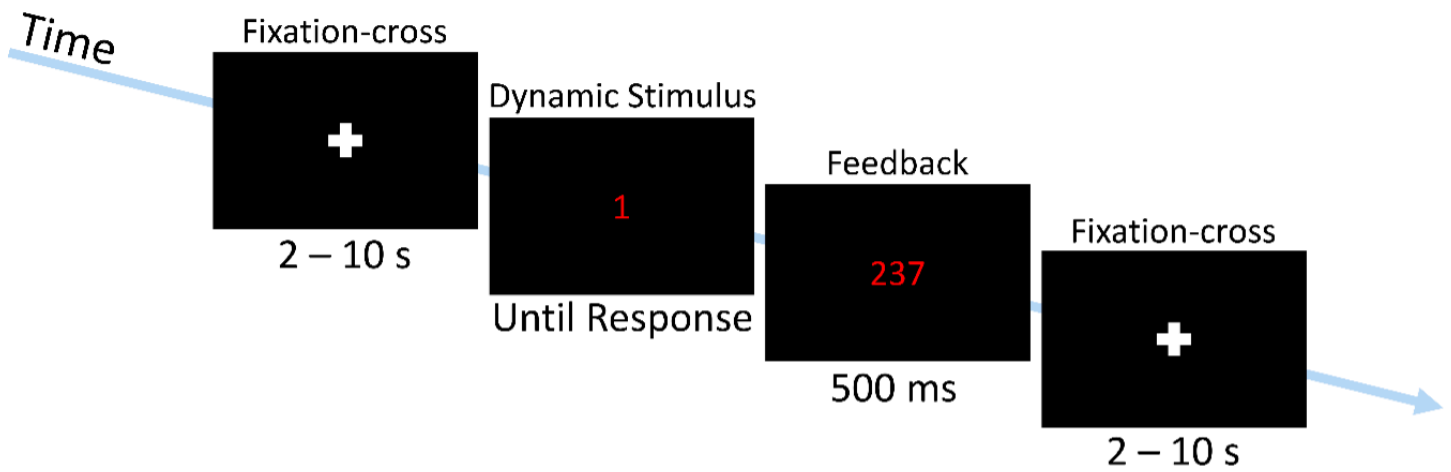


Figure 1

Psychomotor vigilance task schematic. Procedure of the psychomotor vigilance task with stimulus sequence and timing information.

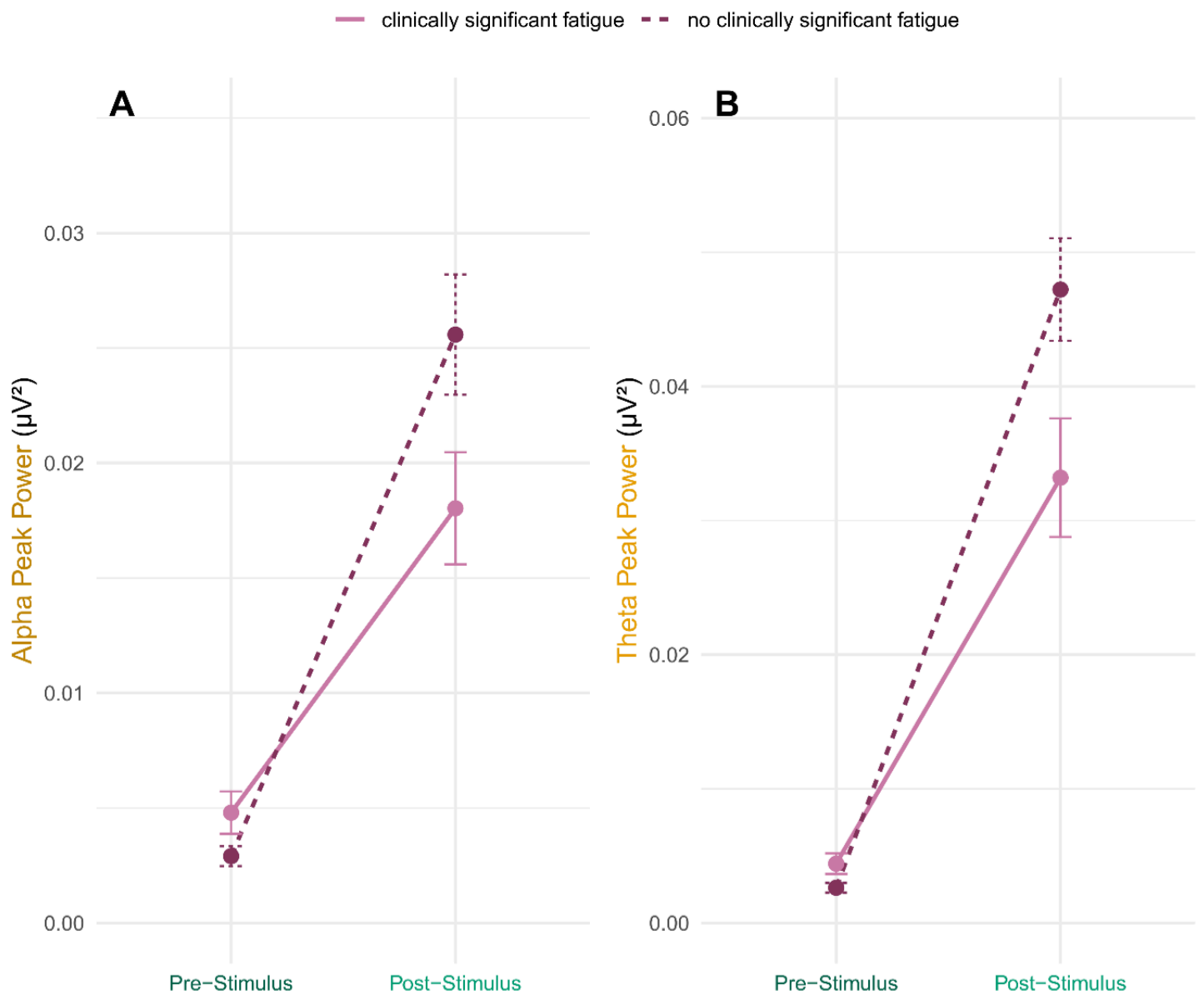


Figure 2

Post-stimulus increases in power spectral density. (A) Alpha (8 - 12 Hz) and (B) Theta (4 - 7 Hz) peak power before and after stimulus onset in the PVT, shown separately for participants with and without clinically significant fatigue. PVT, Psychomotor Vigilance Task.

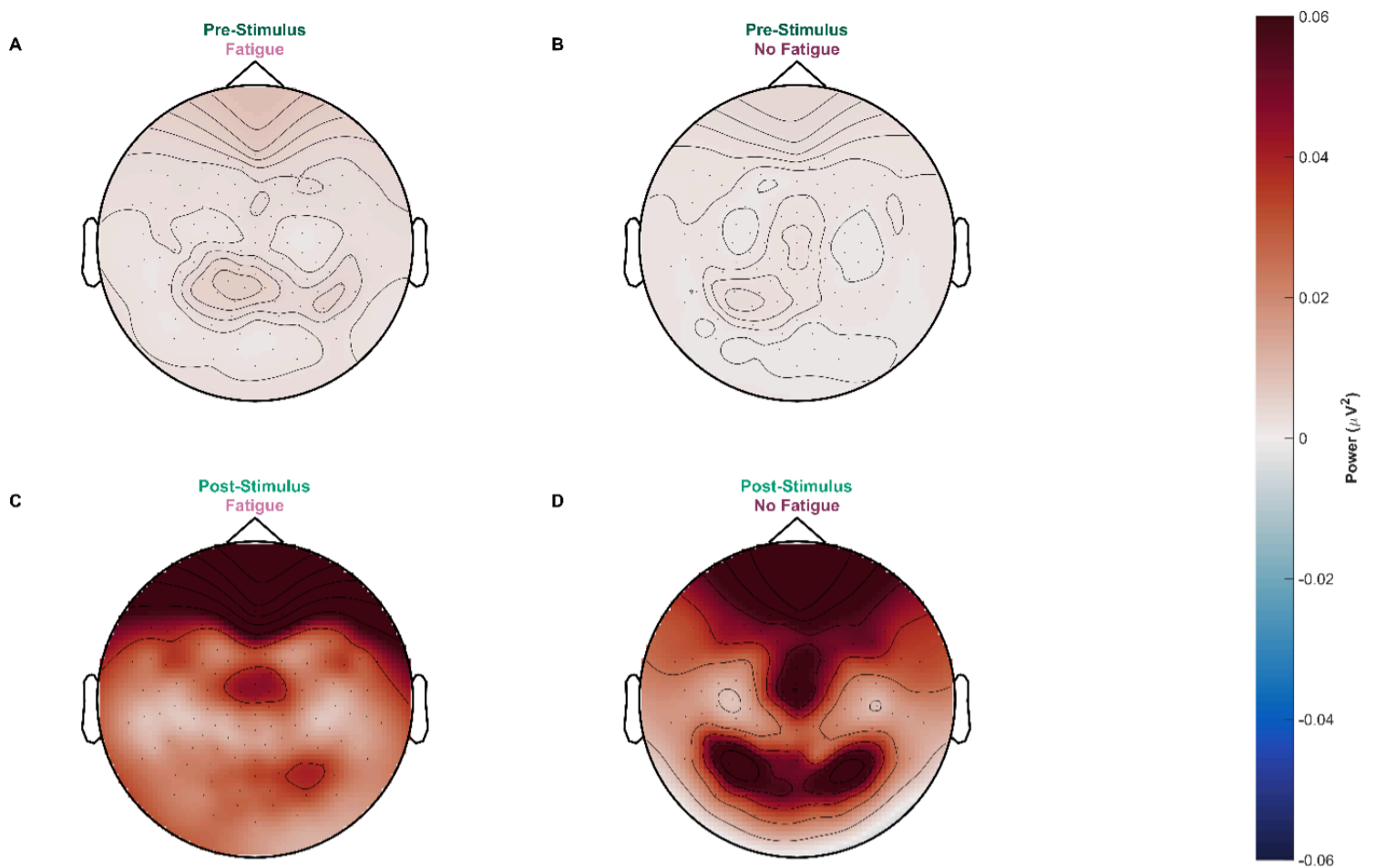


Figure 3

Topographical plots comparing theta power of participants with and without clinically significant fatigue. Topographical plots show theta (4 - 7 Hz) brain activity averaged across all PVT trials and all participants separately for the groups (A, C) with clinically significant fatigue and (B, D) without clinically significant fatigue during the (A, B) pre-stimulus or (C, D) post-stimulus periods. Red areas show an increase in activity. PVT, Psychomotor Vigilance Task.

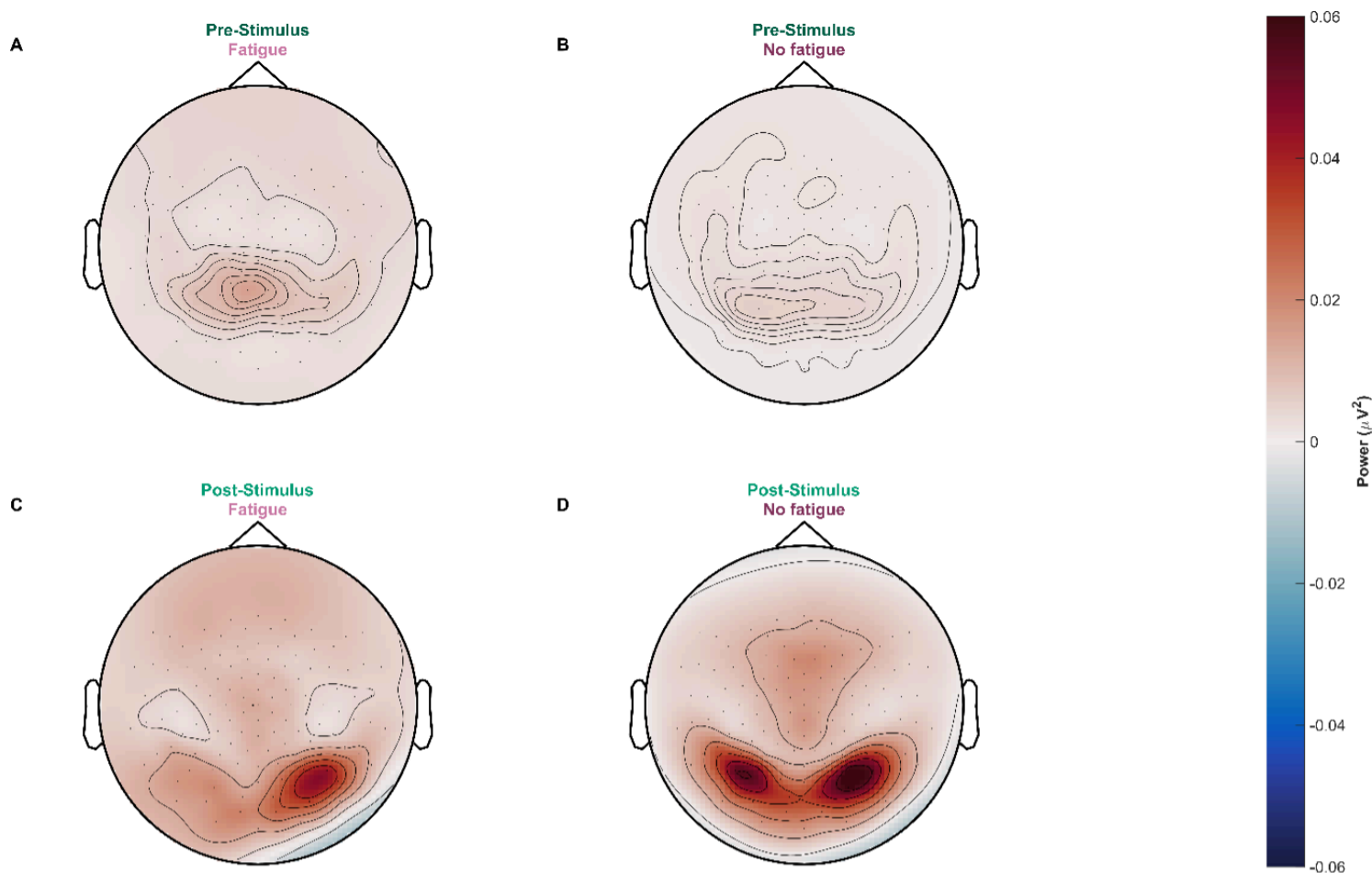


Figure 4

Topographical plots comparing theta power of participants with and without clinically significant fatigue. Topographical plots show alpha (8 - 12 Hz) brain activity averaged across all PVT trials and all participants separately for the groups (A, C) with clinically significant fatigue and (B, D) without clinically significant fatigue, during the (A, B) pre-stimulus or (C, D) post-stimulus periods. Red areas show an increase in activity. PVT, Psychomotor Vigilance Task.

Supplementary Files

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